

Data-Driven and Physics-Based Scaling of Aeroacoustic Noise in Coupled Propeller Systems

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Thesis

Contra-rotating propeller systems have gained increasing attention in UAV applications due to their ability to generate higher thrust within a compact form [1, 2]. In addition, contra-rotating open rotor (CROR) configurations offer improved fuel efficiency and are considered a promising candidate for future commercial aviation [3]. However, their broader adoption is significantly constrained by unfavorable noise characteristics, and existing noise mitigation strategies often come at the expense of aerodynamic performance.

Our previous analyses show that, for such coupled systems, conventional dimensional analysis fails to provide consistent scaling laws. Unlike single-propeller cases, matching dimensionless parameters does not guarantee noise output across scales. To address this, we aim to decompose the underlying noise-generation mechanisms and, in parallel, employ data-driven approaches to construct a scalable mapping between systems of different size.

The test stand is designed to automatically iterate through a wide range of configurations and collect large-scale datasets. These data will support both (1) mathematical modeling of the underlying noise-generation mechanisms, and (2) data-driven approaches, including machine learning methods, to learn the complex relationships between system parameters and noise response.

We believe this will provide a systematic and practical framework for engineers to conduct small scale experiment and optimize such systems, enabling the integration of innovative configurations without compromising performance.

Purpose

Previous studies have explored the scaling method of single-propeller setups by controlling hidden dimensionless parameters such as thrust coefficient, advance ratio, tip Mach number, and tip-path plane angle to assess the scaling effect on BVI noise [4]. After applying this idea into two propeller case following on the physical model introduced by [5], the interaction noise of contra-rotating propellers can be formulated in a dimensionless framework.

In particular, the wake-induced upwash on the downstream blade can be expressed as a superposition of harmonic gusts [6],

$$w = \sum_{n_1=-\infty}^{\infty} w_{n_1} \exp\{ik_X(U_{r2}\tau - X_2)\}, \quad (1)$$

with the streamwise wavenumber

$$k_X = \frac{2n_1 B_1}{DM_{r2}}(M_{T1} + M_{T2}), \quad (2)$$

and reduced frequency

$$\sigma_2 = \frac{k_X c_2}{2}. \quad (3)$$

The unsteady loading is then written as

$$\Delta p = 2\pi\rho_0 U_{r2} w_{n_1} S(\sigma_2, M_{r2}, \bar{X}_2, \Lambda) \exp\left(i\frac{2\pi n_1 \tau}{T}\right), \quad (4)$$

where $S(\cdot)$ is the blade response function. Finally, the far-field acoustic pressure is obtained from [7]

$$p(\mathbf{x}, t) \sim \frac{1}{4\pi c_0} \frac{\partial}{\partial t} \int_{R_h}^{R_t} \int_0^{c_2} \left[\frac{\Delta p(\tilde{R} \cdot \mathbf{n})}{R^*(1 - \mathbf{M} \cdot \tilde{R})} \right]_{\tau=t-R/c_0} dX_2 dr. \quad (5)$$

From these expressions, it is evident that the governing parameters naturally collapse into a set of dimensionless groups, namely

$$M_x, \quad M_T, \quad \frac{c}{D}, \quad \frac{g}{D}, \quad \frac{s}{D}, \quad \frac{r}{D}, \quad B, \quad C_D, \quad \theta, \quad \phi. \quad (6)$$

Therefore, under geometrically similar scaling, the acoustic field can be written in the generic form.

$$p = \rho_0 c_0^2 \Pi\left(\frac{t}{D/c_0}, \frac{R}{D}, M_x, M_T, \frac{c}{D}, \frac{g}{D}, \frac{s}{D}, B, C_D\right), \quad (7)$$

where $\Pi(\cdot)$ is a universal dimensionless function.

Based on this formulation, an ideal scaling strategy would require all these dimensionless quantities to be preserved simultaneously. However, in practical scenarios, only a subset (e.g. M_T , advance ratio, or thrust coefficient) can be matched. To quantify the deviation, we introduce the variation of a generic dimensionless parameter Π_i as

$$\delta\Pi_i = \frac{\Pi_i^{(scaled)} - \Pi_i^{(ref)}}{\Pi_i^{(ref)}}. \quad (8)$$

Because different parameters scale with different powers of D and Ω (e.g. $k_X \sim 1/D$ while $T \sim D$ under fixed M_T), it is generally impossible to keep all Π_i

invariant. As a result, a mismatch is introduced in the reduced frequency σ_2 , the trace Mach number, and the phase function governing the arrival time of acoustic impulses. This mismatch leads to discrepancies in both the amplitude and temporal structure of the radiated noise, even when geometric similarity is strictly enforced.

Our analysis on the data from a collaborating laboratory coincides with this observation, where the measured noise profiles do not follow a same relation under the same Π -based scaling. As shown in Figure 3–5, clear discrepancies remain in both amplitude and gradient characteristics.

However, we observe a consistent similarity in the overall trend of the noise evolution. This suggests that the noise field can be modeled as a vector-valued function of the system scale and governing dimensionless parameters. Let

$\eta \equiv \text{vec}\{\text{NOISE}\}$ then $\eta = \Psi(D, \Pi)$, $\Pi = (\Pi_1, \dots, \Pi_n)^\top$, where D is the characteristic dimension and Π collects the relevant dimensionless groups.

For two geometrically similar systems with characteristic dimensions D_1 and D_2 , the corresponding noise vectors are

$$\eta_1 = \Psi(D_1, \Pi_1), \quad \eta_2 = \Psi(D_2, \Pi_2). \quad (9)$$

If complete similarity were achieved, one would have $\Pi_1 = \Pi_2$; however, the experimental observations indicate that the noise profiles do not collapse under the same Π -based scaling. Therefore, the dependence on scale cannot be eliminated completely, and the mapping must retain an explicit dependence on the system dimensions.

To account for this, we introduce a scale-transfer operator relating the two systems,

$$\eta_2 = \mathcal{T}_{D_1 \rightarrow D_2}[\eta_1]. \quad (10)$$

Ideally, we want to assume a self-similar scaling, the operator would depend only on the scale ratio,

$$\mathcal{T}_{D_1 \rightarrow D_2} = \mathcal{T}_\lambda, \quad \lambda = \frac{D_2}{D_1}. \quad (11)$$

Will we know from the figure, at least the following transformation can be established

$$\mathcal{T}_{D_1 \rightarrow D_2} = \mathcal{T}(D_1, D_2), \quad (12)$$

or, more generally,

$$\eta_2 = \mathcal{T}(D_1, D_2, \Pi_1, \Pi_2)[\eta_1]. \quad (13)$$

Furthermore, when considering coupled systems such as contra-rotating propellers, the radiated noise becomes inherently transient [8]. Additional mechanisms, including potential-field interaction, asymmetric inflow, and tip-vortex interaction, are introduced [9, 10]. In this case, the dominant noise-generation mechanisms vary with the dimensionless spatial coordinates, so that

$$\eta = \Psi\left(D, \frac{x}{D}, \frac{z}{D}, t^*, \Pi\right), \quad t^* = \frac{t}{D/c_0}. \quad (14)$$

Since these mechanisms are organized by the same dimensionless coordinates ($x/D, z/D$) across different scales, their structural dependence is preserved even when exact similarity is not achieved. This indicates that a scale-dependent mapping can still be constructed from a known small-scale response to a large-scale response, thereby making scaling experiments feasible.

Finally, we aim to augment the available dataset to construct a comprehensive, visualization-oriented model of the mapping $\Psi(D, \Pi)$. This will enable the identification of critical points and transition boundaries between different dominant noise-generation mechanisms in the parameter space (e.g. variations in x/D , z/D , and flow conditions). In addition, we will explore data-driven approaches to approximate the underlying mapping. In particular, supervised learning methods such as Gaussian process regression (GPR) [11] for uncertainty-aware interpolation, and neural network based surrogate models (e.g. multilayer perceptrons or physics-informed neural networks, PINNs) [12] will be investigated. Given the observed consistency in the overall trends, these approaches are expected to be particularly promising.

Objective

As a student pilot operating in airports across Southern California, I have observed that many traffic patterns are designed to accommodate local community noise constraints. These constraints often require pilots to deviate from otherwise optimal or intuitive flight paths, introducing additional workload, distraction, and potential safety concerns.

From an engineering perspective, I have also noticed that many innovative aerodynamic and propulsion designs are constrained or even set aside due to noise considerations. This represents a significant limitation, as promising concepts may never be realized in practice.

Motivated by both perspectives, I aim to leverage my background in mathematical modeling to better understand and decouple the underlying mechanisms of noise generation in complex aerodynamic systems. By doing so, I hope to develop a systematic framework that enables designers and engineers to do small scale experiment about their idea more efficiently and be able to incorporate their innovative ideas into aviation systems.

Responsibilities

To achieve these objectives, a fully automated experimental platform is required to support large-scale data acquisition. I am responsible for the CAD design, manufacturing, and assembly of the test stand structure, as well as the design construction of an acoustically isolated enclosure.

In addition, I lead the design, programming, and testing of the control system and data acquisition system. My responsibilities also include data post-processing, documentation, visualization, and, where applicable, the prepara-

tion of research manuscripts.

A prototype of the test stand and partial electronic systems has already been constructed (Fig. 2), and the acoustic enclosure has been completed with the support of my advisor (Fig. 1). The current focus is on transitioning the prototype into a fully operational setup, as well as the development and calibration of a microphone array, as the existing setup is insufficient to accurately capture the spatial and temporal variations of the noise field.



Figure 1: Acoustic enclosure at ASL.



Figure 2: Prototype for electronic and sensor testing.

Once this component is completed, I will integrate the full system, oversee data collection, and carry out subsequent post-processing and model refinement.

Timeline

April 1 – June 10: Complete the physical construction of the test stand and conduct circuit testing. Develop and validate the data acquisition workflow using a secondary microphone array. **June 20 – August 20:** Deploy and calibrate the full system after the arrival of high-fidelity microphones, and carry out large-scale data collection. **After August 20:** Perform data analysis, refine the models, and apply the results to the proposed scaling framework.

Appendix

Figure 3–5 present the measured noise maps at different scales. Each figure corresponds to one scale (propeller diameter). Within each figure, each row represents a reduced vertical separation z/D , and each column represents an RPM setting. For each subpanel, the horizontal axis denotes the reduced axial separation x/D .

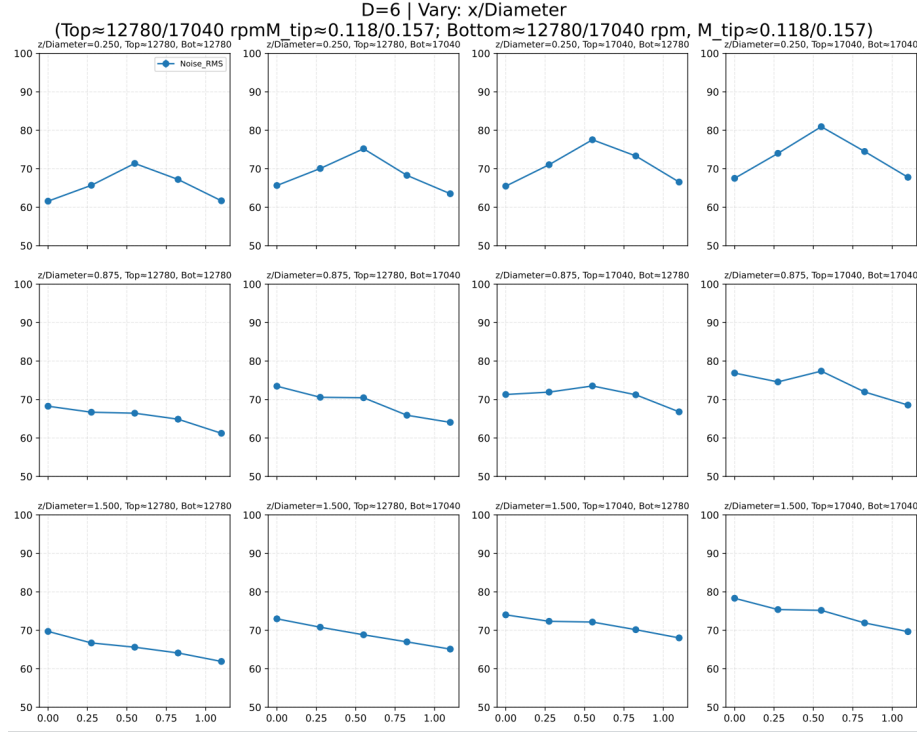


Figure 3: Noise map for the $D = 6$ cm scale. Each row corresponds to a reduced vertical separation z/D , each column corresponds to an RPM setting, and the horizontal axis in each subpanel is the reduced axial separation x/D .

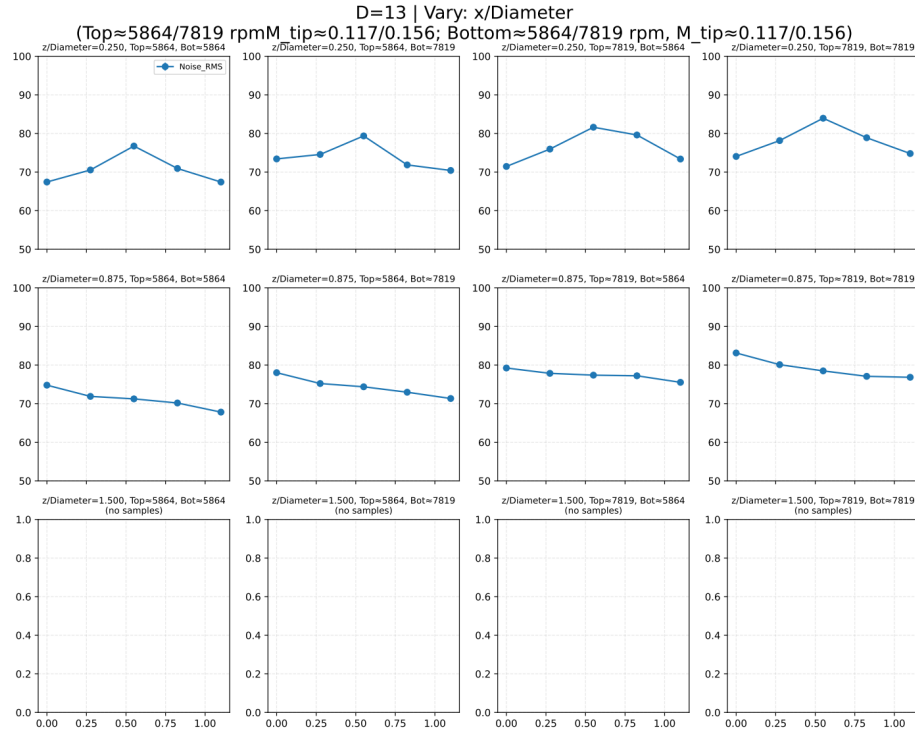


Figure 4: Noise map for the $D = 13$ cm scale. Each row corresponds to a reduced vertical separation z/D , each column corresponds to an RPM setting, and the horizontal axis in each subpanel is the reduced axial separation x/D .

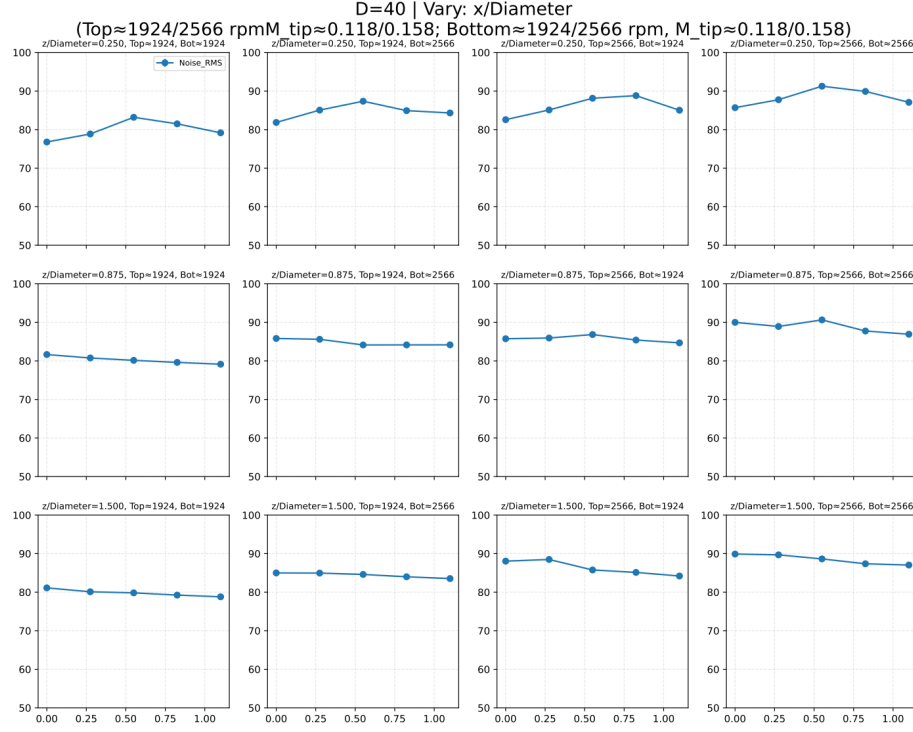


Figure 5: Noise map for the $D = 40$ cm scale. Each row corresponds to a reduced vertical separation z/D , each column corresponds to an RPM setting, and the horizontal axis in each subpanel is the reduced axial separation x/D .

References

- [1] R. S. McKay, M. J. Kingan, S. T. Go, and R. Jung, “Experimental and analytical investigation of contra-rotating multi-rotor uav propeller noise,” *Applied Acoustics*, vol. 177, p. 107850, 2021.
- [2] H. Li, Z. Chen, and H. Jia, “Experimental investigation on hover performance of a ducted coaxial-rotor uav,” *Sensors*, vol. 23, no. 14, p. 6413, 2023.
- [3] I. Stratikis, G. Protopapadakis, P. Katsapoxaki, V. Gkoutzamanis, G. Ghorbaniasl, and A. Kalfas, “Aerodynamic and aeroacoustic sensitivities of contra-rotating open rotors to operational parameters,” *The Aeronautical Journal*, vol. 128, no. 1330, pp. 2832–2855, 2024.
- [4] D. B. Signor, M. E. Watts, F. J. Hernandez, and F. F. Felker, “Blade-vortex interaction noise- a comparison of in-flight, full-scale and small-scale

- measurements,” in *Aeromechanics Specialists Conference on Aerodynamics, Acoustics and Dynamics, San Francisco, CA*, 1994, p. 1.
- [5] M. Kingan and A. Parry, “Time-domain analysis of contra-rotating propeller noise: wake interaction with a downstream propeller blade,” *Journal of Fluid Mechanics*, vol. 901, p. A21, 2020.
 - [6] A. S. Smyth, A. M. Young, and L. Di Mare, “Effect of three-dimensional geometry on harmonic gust–airfoil interaction,” *AIAA Journal*, vol. 59, no. 2, pp. 737–750, 2021.
 - [7] J. E. Ffowcs Williams and D. L. Hawkings, “Sound generation by turbulence and surfaces in arbitrary motion,” *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 264, no. 1151, pp. 321–342, 1969.
 - [8] W. Chen, Z. Ma, É. Spieser, J. Guo, P. Zhou, S. Zhong, X. Zhang, and X. Huang, “Acoustic imaging of the broadband noise arising in contra-rotating co-axial propellers,” *Aerospace Science and Technology*, vol. 141, p. 108531, 2023.
 - [9] C. C. Paruchuri, P. Joseph, D. C. Akiwate, A. B. Parry, and S. D. Prior, “On the noise generation mechanisms of overlapping propellers,” in *AIAA Aviation 2021 Forum*, 2021, p. 2281.
 - [10] R. S. McKay, M. J. Kingan, and R. Go, “Experimental investigation of contra-rotating multi-rotor uav propeller noise,” in *Proceedings of ACOUSTICS*, vol. 10, no. 13, 2019.
 - [11] C. Williams and C. Rasmussen, “Gaussian processes for regression,” *Advances in neural information processing systems*, vol. 8, 1995.
 - [12] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational physics*, vol. 378, pp. 686–707, 2019.